

P-TRANSMIT AI — The AI Model

Operating Manual & Technical Overview

What each layer means and how to run it

Document	AI Model Manual (Manual + Overall Information)
Platform version	0.1.0 — Phase-1 MVP
Date	3 October 2026
Model family	Supervised tabular regression: Random Forest · XGBoost · Ridge
Target	Weekly confirmed malaria cases per region (Ghana, 16 regions)
Forecast horizons	1, 2, 4 and 8 weeks ahead
Validation	Strictly chronological (temporal) hold-out — never random splits
Code reference	<code>backend/app/ml/transmission.py</code> , <code>backend/app/services/alerts.py</code>
Licence	MIT (code) · geoBoundaries ADM1 boundaries CC-BY 4.0

DISCLAIMER — read first. P-TRANSMIT AI is a research and surveillance decision-support system. Model predictions are **not diagnoses** and must not independently determine patient treatment or public-health action. All outputs require review by qualified public-health researchers and officers. Demonstration data are synthetic and always labelled as such.

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Part I — Overall information

What P-TRANSMIT AI is

P-TRANSMIT AI is an open, locally-hostable decision-support platform for *Plasmodium falciparum* transmission, genomic epidemiology and antimalarial drug resistance surveillance in Ghana. It ingests weekly surveillance data (cases plus environmental covariates), trains and compares supervised forecasting models, produces per-region forecasts with uncertainty ranges, explains what drove each prediction, and raises relative early-warning signals for human review.

Module	Status	What it does
Dashboard	Live	National KPIs, choropleth map, national forecast, per-region explanations, alerts.
Transmission AI	Live	Trains models, shows held-out observed-vs-predicted curves and metrics per horizon.
Spatial Intelligence	Live	Interactive map of Ghana's 16 regions with risk / observed / forecast layers.
Forecasting	Live	1/2/4/8-week forecasts with residual-based uncertainty intervals and stated assumptions.
Explainable AI	Live	SHAP (when installed) or perturbation-based per-region contributions — model behaviour, not causality.
Early-warning alerts	Live	Level / trend / forecast signals relative to the dataset's own history; human review required.
Data Explorer	Live	CSV/TSV/XLSX/JSON upload, data-quality report, column mapping, Ghana region fuzzy-matching.
Model Laboratory	Live	Train/compare/manage models with full reproducibility provenance.
Reports	Live	Markdown research report including a limitations section.
Plasmodium Genomics / Resistance	Scaffold	Marker catalogue + descriptive statistics; awaits validated data (no simulated results).

Scientific governance principles (enforced in code)

- **Synthetic data are always labelled** and never mixed with real data.
- **No invented thresholds.** Risk categories are relative to the loaded dataset's own history; resistance interpretation requires researcher-supplied definitions.
- **Strictly temporal validation.** Time-series rows are never shuffled; train weeks always precede test weeks.
- **MAE / RMSE are primary metrics** for case counts; R^2 is reported for completeness only.
- **Explanations describe model behaviour, not causal effects.**
- **Alerts are research signals, not confirmed outbreaks;** they require human and public-health review.
- **Genetic marker $\neq$ phenotypic drug resistance.**
- **Audit logs** record logins, dataset operations, training runs and report generation.

Technology stack

Layer	Technology
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Backend / API	Python · FastAPI · SQLAlchemy (SQLite default, PostgreSQL optional) · scikit-learn · XGBoost · pandas / numpy
Frontend	React · TypeScript · Vite · Recharts (charts) · Leaflet (map)
Geography	geoBoundaries gbOpen GHA ADM1 (16 regions), CC-BY 4.0, simplified for the web map
Deployment	Local (one-click .bat) or one Docker image serving UI + API together

Part II — The AI model: what each layer means

The "AI model" is not a single neural network — it is a **pipeline of ten layers**. Each layer has one job; together they turn raw weekly surveillance rows into forecasts, explanations and early-warning signals.

1 • Data — standardised weekly rows	2 • Features — lags, environment, season, region	3 • Models — RF · XGBoost · Ridge
4 • Validation — chronological split	5 • Metrics — MAE / RMSE primary	6 • Forecast — next weeks per region
7 • Uncertainty — residual-based intervals	8 • Risk — relative categories	9 • Explainability — SHAP / perturbation
10 • Alerts — relative early-warning rules		

Layer 1

Data layer — the raw material

What it is: a weekly surveillance table, one row per region per epi-week.

- Required columns: `region` , `week_start_date` , `malaria_cases` (synonyms auto-detected, e.g. `mal_cases` , `cases` , `date`).
- Optional covariates: `rainfall_mm` , `temperature_mean_c` , `humidity_pct` , `ndvi` , `population_illustrative` , `suspected_cases` , `test_positivity_rate` .
- Ingestion profiles data quality, suggests column mappings and fuzzy-matches names to Ghana's 16 canonical regions. **Nothing is imputed silently** — missing values are reported.

What it means: garbage in → garbage out. The Data Explorer's quality report is your first checkpoint before training.

Layer 2 Feature layer — what the model actually sees

For every region-week, the engine (`build_supervised`) constructs:

Feature group	Columns	Meaning
Case history (lags)	<code>cases_lag1...cases_lag4</code>	Confirmed cases in each of the previous 1–4 weeks (per region). The dominant predictor: recent momentum.
Environment (current + lag)	<code>rainfall_mm</code> , <code>temperature_mean_c</code> , <code>humidity_pct</code> , <code>ndvi</code> (+ <code>_lag1</code> of each)	Weather/vegetation conditions now and one week ago — known correlates of transmission.
Seasonality	<code>sin_week</code> , <code>cos_week</code>	Cyclic encoding of the ISO week-of-year (period 52) so the model can learn wet/dry-season patterns without a discontinuity at year boundaries.
Region fixed effects	<code>region_...</code> dummies (one per region)	One-hot indicator for the row's region — lets a <i>single global model</i> capture each region's baseline level.
Population (optional)	<code>population_illustrative</code>	Denominator/weight column when present.
Target	<code>target_lead</code> = cases at week $t+h$	For horizon h , the label is the case count h weeks in the future (a lead shift, <code>shift(-h)</code>).

What it means: the model predicts the future from (a) recent case history, (b) current environment, (c) time of year, and (d) which region it is. Rows with missing lag/lead values are dropped — that is why the first weeks of each region are not used.

Layer 3 Model layer — three competing algorithms

Each horizon trains **three** candidate models (`DEFAULT_MODELS`); the platform keeps the best one:

Model	Hyper-parameters (fixed in v0.1)	Plain-language explanation
Ridge	<code>StandardScaler</code> → <code>Ridge(α = 1.0)</code>	A regularised <i>linear</i> model. Learns one weight per feature: stable, fast, fully interpretable, but only captures linear relationships. Serves as the honest baseline.
Random Forest	250 trees, <code>min_samples_leaf</code> = 2 , seed fixed	Averaged ensemble of decision trees ("bagging"). Captures non-linearity and interactions (e.g. rainfall × season × region), robust to outliers, rarely overfits hard. Slower and less interpretable than Ridge.
XGBoost	500 trees, learning-rate 0.05, depth 4, subsample 0.9, colsample 0.9, λ 1.0	Gradient <i>boosting</i> : trees added sequentially, each correcting the previous one's errors. Usually the strongest on tabular data; the most sensitive to settings and to data quality.

- Training happens **once per horizon** (1, 2, 4, 8 weeks) — so up to $4 \times 3 = 12$ fits per run.
- The winner per horizon = **lowest test-set MAE**. It is persisted as a joblib artifact, e.g. `transmission_xgboost_h4_ds1.joblib` , together with its full provenance (feature list, train/test weeks, seed, software versions, timestamp).
- A fixed **random seed** (default 42, user-settable) makes runs reproducible.

What it means: you don't pick the algorithm — you offer candidates and the platform picks the most accurate on held-out future weeks, per horizon, and records exactly how.

Layer 4 Validation layer — honest evaluation

- **Chronological split on unique weeks** (`temporal_split`): the earliest **80%** of weeks (configurable "train fraction") form the training set; the latest 20% form the test set.
- Train-week and test-week sets are *disjoint* and `max(train week) < min(test week)` — verified by the automated test suite. No future information can leak backwards.
- Lags are computed within each region's own history before the split, so no row borrows information from a test week.

What it means: reported metrics come from genuinely unseen future weeks — the score approximates how the model would have behaved in production, not how well it memorised the past.

Layer 5 Metrics layer — how good is good?

Metric	Formula / rule	How to read it
MAE (primary)	mean predicted – observed	"Typical error in cases per week." Compare it to your region's case scale: MAE 8 on counts of ~100 ≈ 8% typical miss.
RMSE (primary)	$\sqrt{\text{mean}(\text{error}^2)}$	Like MAE but punishes large misses heavily — a big gap between RMSE and MAE signals occasional bad predictions.
MAPE	mean err/observed on non-zero weeks	Percentage error; unstable when counts are near zero, hence excluded on zero weeks.
sMAPE	symmetric % variant	Bounded percentage variant for comparison across regions.
R ²	coefficient of determination	Reported <i>for completeness only</i> ; low R ² is normal and not alarming for noisy count data.
residual_std	std of test residuals	Feeds the uncertainty intervals (Layer 7).
n_test, fit_seconds	—	How many held-out rows were scored; how long fitting took.

What it means: judge models by **MAE/RMSE in case units**, per horizon — longer horizons (8 weeks) are inherently harder; expect their errors to be larger.

Layer 6 Forecast layer — producing next weeks' numbers

- At the **last observed week**, the engine builds one feature row per region (lags from actual history, seasonality from the calendar, region dummy = 1).
- Each persisted best model predicts the case count h weeks ahead ($h \in \{1, 2, 4, 8\}$).
- Environmental persistence assumption:** future weather/vegetation are held at their *last observed value*. The API states this in every forecast response (`assumptions` field) — the model does not know next month's rainfall.
- Predictions are clipped at ≥ 0 (case counts cannot be negative).

What it means: a forecast is "the best guess given conditions stay roughly as they are" — read it alongside the stated assumptions, not as a certainty.

Layer 7 Uncertainty layer — the error bars

- Each forecast carries `interval_low` / `interval_high` = prediction $\pm 1.96 \times \text{residual_std}$ (low clipped at 0).
- `residual_std` is the standard deviation of the model's errors on the held-out test weeks — i.e. "how wrong this model has actually been on unseen data."
- Method string returned with every forecast: *"residual-based Gaussian approximation (not calibrated probabilistic intervals)."*

What it means: the interval shows the model's typical error range — it is *not* a 95% probability guarantee, and it will be wide where the model is shaky. Treat values near the upper bound as "worth watching."

Layer 8 Risk layer — relative categories

Context	Rule (computed from the loaded dataset itself)
Forecast rows	predicted $\geq$ historical p90 $\rightarrow$ <i>very high</i> $\cdot$ $\geq$ p75 $\rightarrow$ <i>high</i> $\cdot$ $\geq$ p50 $\rightarrow$ <i>moderate</i> $\cdot$ else <i>low</i> (quantiles of all weekly cases in the dataset).
Map "risk" layer	Per region, relative to <i>that region's own</i> historical p90: $\geq$ p90 $\rightarrow$ very high $\cdot$ $\geq$ 0.9·p90 $\rightarrow$ high $\cdot$ $\geq$ 0.75·p90 $\rightarrow$ moderate $\cdot$ else low.

What it means: categories answer "*unusual for this data?*" — they are **not** national clinical thresholds. A "very high" in a chronically low region may still be a small number; always read it with the case values next to it.

Layer 9 Explainability layer — "why this number?"

- **Global importance:** permutation importance on the held-out weeks (5 repeats, scored by MAE change) — which features the model relies on overall; shown as top-12.
- **Per-region explanations** (Dashboard "Why?" panel): the engine perturbs each non-region feature (replaces it with its training median) and records how the prediction moves; where `shap` is installed, tree models use **SHAP TreeExplainer** instead. Top 8 contributions per region are shown as signed bars.
- Positive contribution = pushed the prediction *up*; negative = pulled it *down*.
- Method text always ends with: "*model behaviour, not causal effects.*"

What it means: explanations tell you what the model leaned on — useful for sanity-checking ("is it reacting to rainfall or to last week's spike?") — but they are **not** evidence that feature *caused* transmission changes.

Layer 10 Alert layer — relative early-warning rules

Three deterministic rules scan every region (minimum 30 weeks of history) — see `alerts.py` :

Rule	Severity	Trigger
Level	warning	Mean of the last 4 observed weeks $>$ that region's historical 90th percentile .
Trend	warning	Least-squares slope of the last 8 weeks $>$ the 90th percentile of all historical 8-week slopes for that region (and threshold $>$ 0).
Forecast	info	Median model prediction for weeks 1–4 $>$ the region's historical p90.

- Every alert stores its **evidence** (the numbers that fired it), the model reference and a plain-language explanation.
- Duplicates are suppressed: an open alert of the same type+region is not re-created on the next scan.
- Workflow: *open* $\rightarrow$ *acknowledged* / *dismissed* — a person always decides.

What it means: alerts are "unusual relative to this dataset" flags for triage — never confirmation of an outbreak, and never a treatment directive.

Part III — Operating the model (step by step)

A. Access the platform

- 1 **Local:** double-click `start-pttransmit.bat` → open `http://localhost:4173` (both server windows must stay open; close them to stop).
Hosted: open your permanent deployment URL (e.g. the Render link) — the same UI and API are served from one address.
- 2 Sign in: click ⚡ **Enter live demo** (read-only researcher, no account), or use `demo / demo1234` or `admin / pttransmit-admin`. Change real credentials before any real deployment.

B. Train the models (Model Laboratory)

- 1 Open **Data Explorer** and confirm your dataset is listed (the clearly-labelled DEMO dataset loads automatically in demo mode).
- 2 Open **Model Laboratory** and set:
 - **Dataset** — the surveillance table to learn from.
 - **Random seed** (default 42) — same seed ⇒ same results.
 - **Train fraction (temporal)** (default 0.8) — earliest 80% of weeks train, latest 20% test.
 - **Models** — tick Random Forest, XGBoost, Ridge (all three by default).
 - **Horizons (weeks)** — 1 / 2 / 4 / 8 (all by default).
- 3 Press **Train** and wait (~1 minute on a laptop for the demo dataset; longer on free-host CPUs). Training continues per horizon even if one model errors.
- 4 Read the results table: for each horizon you get every candidate's MAE/RMSE/R², the **best model** (lowest test MAE) and its saved artifact path. Expand a row's **Provenance** for dataset, features, seed, versions and timestamp.

Training has run once already? The **Transmission AI** page has the same train button plus the region/model selectors for inspecting held-out curves — useful for a second opinion before trusting a horizon.

C. View forecasts

- 1 **Dashboard:** KPI cards (last week, recent 4-week national cases, *national forecast next week*, regions at elevated risk) and the map. Use the **Map layer** selector: *Transmission risk category / Recent 4-week case mean / Forecast median (≤4 weeks)*. Hover any region for detail.
- 2 **Spatial Intelligence:** the same map with a snapshot table of all regions (observed, historical p90, forecast, risk).
- 3 **Transmission AI:** per region and per model, held-out observed-vs-predicted curves and the metric block — check these before quoting any forecast.

D. Read the explanations

- 1 Dashboard → **"Why? Top contributors for (region)"**: signed bars of the top features behind that region's current prediction, with the method note above them. Ask: *is the model leaning on recent cases, rainfall, or season?*

E. Run and triage alerts

- 1 Open **Alerts** → **Run alert scan**. New signals appear with severity, region, evidence numbers and the explanation text.
- 2 Review each signal against your operational knowledge, then **Acknowledge** (keeping it on the radar) or **Dismiss** (not actionable). Already-open signals of the same type+region are not duplicated on re-scan.

F. Load real data & retrain

- 1 **Data Explorer** → choose record type *surveillance* → upload CSV/XLSX/JSON with `region`, `week_start_date`, `malaria_cases` (synonyms auto-detected). Review the data-quality report and column mapping; confirm region matches.
- 2 **Model Laboratory** → select the *new* dataset → Train. Dashboard, forecasts, explanations and alerts now default to the latest dataset.
- 3 Regenerate a **Report** for the record (Markdown, includes limitations), and check **Settings** for demo-mode status and the governance checklist.

Housekeeping on free/hosted tiers: storage is ephemeral — after a restart the labelled demo data re-seeds automatically, but *trained model files and uploaded accounts do not survive*. Re-train via Model Laboratory after every redeploy. Locally, artifacts live in `models/` and persist across restarts.

Part IV — Interpreting outputs & limitations

You see	Read it as	Do NOT read it as
MAE = 8 (h=1)	On unseen future weeks the typical miss was ± 8 cases/week.	"8% error" or a guarantee for any single week.
RMSE $\gg$ MAE	Occasional large misses — inspect the observed-vs-predicted curve for spike weeks.	A broken model (spikes are inherently hard).
R ² low or negative	Common for noisy count series; MAE/RMSE are primary.	The model is useless — compare against the baseline.
Interval [120 – 340]	Typical model error band from held-out residuals.	A calibrated 95% probability range.
Risk: "very high"	Prediction is $\geq$ the dataset's own historical p90.	A national emergency threshold.
Top contributor: rainfall +45	The model leaned on rainfall for this prediction.	Rainfall <i>caused</i> the increase.
Level alert fired	Last 4 weeks unusually high <i>for this region's own history</i> — verify.	A confirmed outbreak.

Known limitations (v0.1)

- **Environmental persistence:** forecasts hold weather/vegetation at the last observed value; genuine rainfall shocks will not be anticipated.
- **Direct (non-recursive) horizons:** each horizon is a separate model; an 8-week forecast does not internally compound the 1–4 week ones.
- **Intervals are uncalibrated** Gaussian approximations from test residuals.
- **Regional dummy coverage:** a region absent at training time gets no dedicated dummy — add history for all 16 regions before real use.
- **Minimum history:** alert rules need ≥ 30 weeks; lag construction consumes the first 4+ weeks per region.
- **Demo data are synthetic** (with a scripted late-season upswing to exercise alerts) and always labelled.
- **Genomics/Resistance modules are descriptive scaffolds** awaiting validated data; a detected marker is not phenotypic resistance.

Bottom line: P-TRANSMIT AI supports research and surveillance judgement. Predictions are not diagnoses and must not independently determine treatment or public-health action.

Quick troubleshooting

Symptom	Fix
Dashboard blank / no forecast	No models trained yet → Model Laboratory → Train . (A crash boundary now shows an inline error instead of a blank page.)
Forecast column shows "—"	Same as above; also check the selected dataset matches the trained artifact (ds id).
First page load takes ~50 s	Free-tier instance was asleep — first request wakes it; subsequent loads are fast.
Alerts scan returns nothing	Signals are relative and rare by design; ensure ≥ 30 weeks of history and that open alerts already cover the region/type.

Accounts/data gone after redeploy	Ephemeral storage — demo data re-seeds; re-train models and re-upload files.
Upload rejected	Surveillance uploads need <code>region</code> + <code>week_start_date</code> + <code>malaria_cases</code> (synonyms accepted); check the quality report for mapping issues.

Appendix

A. Anatomy of a trained artifact

File: `models/transmission_<type>_h<horizon>_ds<dataset_id>.joblib` containing:

Field	Contents
<code>model</code>	The fitted scikit-learn / XGBoost estimator.
<code>model_type</code> , <code>horizon</code>	Which algorithm and lead time it serves.
<code>feature_names</code>	Exact column order the model expects (lags, environment, season, region dummies).
<code>metrics</code>	MAE, RMSE, MAPE, sMAPE, R^2 , <code>residual_std</code> , <code>n_test</code> , <code>fit_seconds</code> .
<code>feature_importance</code>	Permutation-importance ranking on held-out weeks.
<code>train_weeks</code> , <code>test_weeks</code>	The precise chronological split used — reproducibility proof.
<code>seed</code> , <code>software</code> , <code>trained_at</code>	Random seed, Python/scikit-learn/XGBoost/pandas/numpy versions, UTC timestamp.

B. Forecast row fields

`region`, `horizon_weeks`, `last_observed_week`, `forecast_week`, `predicted_cases` (clipped ≥ 0), `interval_low` / `interval_high` ($\pm 1.96 \cdot \text{residual_std}$), `risk_category`, `model_type`, `interval_method` — plus the response-level `assumptions` string (environmental persistence).

C. Glossary

Term	Meaning
Horizon (h)	How many weeks ahead are being predicted (1, 2, 4, 8).
Lag	A past value of the same series (e.g. cases one week ago).
Lead / <code>target_lead</code>	The future value being predicted (shift $-h$).
Temporal split	Train on earlier weeks, test on later weeks — no shuffle, no leakage.
Quantile (p50/p75/p90)	The value below which 50/75/90% of historical observations fall — the basis of relative risk bands.
Permutation importance	How much held-out accuracy drops when a feature's values are shuffled — a model-agnostic "who matters" ranking.
SHAP	Game-theory-based feature attribution for tree models; used when the optional <code>shap</code> package is installed.
Residual	Prediction – observed on held-out weeks; <code>residual_std</code> sizes the intervals.
Provenance	The recorded context of a training run (data, split, seed, versions) that makes it reproducible.